

⊕ Hopf-Cole gives mathematical route to a soln for Burgers. Date: / /

⊕ A numerical method based on Crank-Nicolson scheme for Burgers' eqn

↳ state that Hopf-Cole transforms Burgers' eqn into linear heat eqn, after which they solve the heat eqn & recover Burgers' soln

⊙ Hopf-Cole Math

$$u_t + uu_x = \nu u_{xx}$$

→ papers $\Rightarrow$ $u_t + uu_x = \boxed{\begin{matrix} 1 \\ \text{Re} \end{matrix}} u_{xx}$

↳ Say, burgers is simplified form of Nav-Stokes $\nu = \frac{1}{\text{Re}}$

⊕ $\gamma u = 3 \Rightarrow uu_x = 3u_x$ as $uu_x = 10u_x$

uu_x

⊕ uu_x multiplied by own derivative
↳ non linear

$u \& u_x$
↳ both
linear

$$u_t + uu_x = \nu u_{xx}$$

↳ $\phi_t = \nu \phi_{xx}$ — heat eqn

⊕ uu_x — disappears

⊕ uu_x is no product of u & u_x — disappears
own deriv.

① heat eqⁿ is linear in ϕ

How? define a new function

$$\hookrightarrow \phi(x, t)$$

& u relate to ϕ

Hopf-Cole transformation

$$u = -\frac{2}{Re} \frac{\phi_x}{\phi}$$

$$\text{or } \Rightarrow u = -\frac{2}{Re} \frac{\partial \phi / \partial x}{\phi}$$

$\frac{\phi_x}{\phi} \rightarrow$ spatial rate of change div. by ϕ

② instd of solving dir. for u , let's represent u using another function ϕ .

similar to

$$y = \frac{g'(x)}{g(x)}$$

$\rightarrow$ rather than solving for y we're changing the repr. of solⁿ

③ if we substitute $\Rightarrow u = -\frac{2}{Re} \frac{\phi_x}{\phi}$

$\hookrightarrow$ into Burgers eqⁿ

we get $\Rightarrow \phi_t = \frac{1}{Re} \phi_{xx} \hookrightarrow$ linear heat eqⁿ

$$u_x + uu_x = (1/Re) u_{xx}$$

$\downarrow$ Hopf-Cole trans.

heat eqⁿ

$$\phi_t = (1/Re) \phi_{xx}$$

we are given solⁿ for u to solve for ϕ

Then once we know ϕ , we recover u .

$$u = -\frac{2}{Re} \frac{\phi_x}{\phi}$$

$$\text{So: } \phi \rightarrow u$$

④ what IC?

$\hookrightarrow$ Let's say Burgers $\Rightarrow u(x, 0) = u_0(x)$

$\hookrightarrow$ need to def. what the corr. IC for ϕ should be:

paper shows

$$\phi(x, 0) = \exp \left[-\frac{Re}{2} \int_0^x u_0(\eta) d\eta \right]$$

Date: / /

Burgers / $u(x,0) = u_0(x)$ Heat eqn IC

Hopf-cole

$$\phi(x,0) = \exp\left[\frac{-Re}{2} \int_0^x u_0(\eta) d\eta\right]$$

(i) translating into ϕ problem

co.b (BC)

after transform $\Rightarrow \phi(x(0,t) = 0$ &

$$\phi(x(1,t) = 0$$

(ii) Dirichlet to Neumann

$$u(0,t) = 0 \Rightarrow \phi_x(0,t) = 0$$

We start with

nonlinear Burgers eqn
Hopf-cole

linear heat eqn

Paper for Solving heat eqn using Fourier series

So we basically need a repn which used Hopf-cole method to obtain values

Conceptually $\Rightarrow \phi(x,t) = A_0 + \sum_{n=1}^{\infty} A_n e^{-n^2 \pi^2 t / Re} \cos(n\pi x)$

the heat eqn can be solved mathematically in terms of known basis functions

then $\Rightarrow \phi \rightarrow u$ w/ Hopf-cole

$\therefore$ analytical/exact soln of Burgers for IC & BC is obtain

(*) PINN $\Rightarrow u_0(x,t)$ & transform w/ PDE + IC + BC we don't give it exact $u(x,t)$

(*) Hopf-cole $\Rightarrow u_{ref}(x,t)$ is obtained mathematically

then compare then $\Rightarrow u_0(x,t)$ v.s. $u_{ref}(x,t)$

math. origin of ref. soln

This paper $\Rightarrow$ Hopf-cole $\rightarrow$ heat $\rightarrow$ (math) $\rightarrow$ discrete num. soln
can Nielsen

(*) Does PINN req. to become linear?

Date: / /

can turn on

(NO) $\rightarrow U_t + U_n - V_{unn} \rightarrow$ very useful

① we don't have to perform
Hopt-calc first

network independently calculate

U_a, U_n, U_t, U_{unn}

Hopt-calc $\Rightarrow$ answer $\rightarrow$ where does trusted Burger,
net. soln come from

② Derivation

Why $\Rightarrow$ $U = -\frac{2}{Re} \frac{\phi_n}{\phi}$

$U_t + U_n = \frac{1}{Re} U_{unn} \rightarrow$ let,
 $v = \frac{1}{Re}$

$U_t + U_n = v U_{unn}$

(Hopt-calc)

difficult part

$U = -2v \frac{\phi_n}{\phi}$ (Representation)

non linear
term
yet
calculated
then log.
der.
structure

Date: / /

$\frac{\phi_n}{\phi} \Rightarrow \frac{d}{dn} \ln \phi = \frac{\phi_n}{\phi}$

$\therefore U = -2v (\ln \phi)_n$ force come

Burger $\rightarrow U_t, U_n, U_{unn}$
so if $U = -2v (\ln \phi)_n$

We need to calc $\Rightarrow$ let
all three $\eta = \ln \phi$

then $\Rightarrow U = -2v \eta_n$

Calc $\Rightarrow U = -2v \eta_n$
 $U_t = -2v \eta_{nt}$ (diff w.r. time)

$U_n = -2v \eta_{nn}$

$U_{unn} = -2v \eta_{nnn}$

Assume $\Rightarrow U U_n$

sub $\Rightarrow U = -2v \eta_n$

& $U_n = -2v \eta_{nn}$

$U U_n = (-2v \eta_n)(-2v \eta_{nn})$

Date: / /

$$\therefore u_{nn} = 4v^2 q_n q_{nn} \quad \text{non linear term}$$

put into burgers:

$$u_t + u_{nn} = 2u_{nn}$$

$$-2v q_{nt} + 4v^2 q_n q_{nn} = v(-2v q_{nnn})$$

$$-2v q_{nt} + 4v^2 q_n q_{nn} = -2v^2 q_{nnn}$$

$$\text{Div by } -2v \Rightarrow \boxed{q_{nt} - 2v^2 q_n q_{nn} = v q_{nnn}}$$

doesn't look
like tho

$$\Rightarrow q = \ln \phi$$

$$q_t = \frac{\phi_t}{\phi}$$

$$2 \quad q_n = \frac{\phi_n}{\phi}$$

$$\text{Diff } q_n \Rightarrow \boxed{q_{nn} = \frac{\phi_{nn}}{\phi} - \frac{\phi_n^2}{\phi^2}}$$

Date: / /

$$\Rightarrow \boxed{q_{nn} = \frac{\phi_{nn}}{\phi} - \frac{\phi_n^2}{\phi^2}}$$

$$\text{let } F = q_t - v q_{nn} + v q_n^2$$

$$\text{when } F_n = q_{tn} - v q_{nnn} + 2v q_n q_{nn}$$

$$\text{transform burgers eq} \rightarrow q_{nt} - 2v q_n q_{nn} = v q_{nnn}$$

$$\textcircled{1} \quad q_{nt} - v q_{nnn} + 2v q_n q_{nn} = 0$$

$$\boxed{F_n = 0} \quad \textcircled{2} \quad F \text{ is independent of } n$$

$$\text{as } q = \ln \phi, \quad q_t = \frac{\phi_t}{\phi}, \quad q_{nn} = \frac{\phi_{nn}}{\phi} - \frac{\phi_n^2}{\phi^2}$$

$$\Delta 1 v \Rightarrow q_n^2 = \frac{\phi_n^2}{\phi^2}$$

$$\text{so } \Rightarrow q_t - v q_{nn} + v q_n^2$$

$$\text{becom } \Rightarrow \frac{\phi_t}{\phi} - v \left(\frac{\phi_{nn}}{\phi} - \frac{\phi_n^2}{\phi^2} \right) + v \frac{\phi_n^2}{\phi^2}$$

$$\hookrightarrow +v \frac{\phi_n^2}{\phi^2} - v \frac{\phi_n^2}{\phi^2} = 0$$

Date: / /

$$\frac{\phi_t}{\phi} - \gamma \frac{\phi_{xx}}{\phi} - \gamma \frac{\phi^2_{xx}}{\phi^2} + \gamma \frac{\phi^2_x}{\phi^2}$$

$$\left| \frac{\phi_t}{\phi} - \gamma \frac{\phi_{xx}}{\phi} = 0 \right|$$

multi. by ϕ : $\phi_t - \gamma \phi_{xx} = 0$

$$\phi_t = \gamma \phi_{xx}$$

Heat eqn

Paper solved with proceeds to a
Fourier-Series soln

Solve $\Rightarrow \phi_t = \gamma \phi_{xx} \rightarrow$ put in
back into $\left| u = -2\gamma \frac{\phi_{xx}}{\phi} \right|$

$$\therefore \gamma = \frac{1}{Re}$$

$$\left| u = -\frac{2}{Re} \frac{\phi_{xx}}{\phi} \right|$$

exact trans-
form in
the paper



This is how ref soln was
formed

Date: / /

Burgers $\rightarrow$ Hopf-Cole $\rightarrow$ Heat $\rightarrow \phi \rightarrow$ Unref

Finally $\Rightarrow$ Heat eqn $\rightarrow$ ref. Burgers soln

$$\phi_t = \frac{1}{Re} \phi_{xx}$$

Got from Hopf-Cole
& solved w/ Fourier
series

N.B.C. $\Rightarrow \phi_n(0,t) = 0, \phi_n(1,t) = 0$

I.C. $\Rightarrow \phi(n,0) = \phi_0(n)$

Now $\Rightarrow$ we got $\phi(n,t)$ that satisfies ~~heat eqn~~ heat eqn,
I.C. & B.C

$\hookrightarrow$ linear PDE problem

Fourier: Complicated f^n can be repre. as combi.
of simpler funth

(e.g.) $f(x) = A_0 + A_1 \cos(\pi x) + A_2 \cos(2\pi x) + \dots$
 $\hookrightarrow$ Fourier series

Take $\Rightarrow \cos(n\pi x)$

a.e. $\left(\frac{d}{dx} \cos(n\pi x) = -n\pi \sin(n\pi x) \right)$

at $x=0$: $\sin(0) = 0$, At $x=1 \Rightarrow \sin(n\pi) = 0$

$$\therefore \phi_n(0,t) = \phi_n(1,t) = 0$$

② cosine functions naturally satisfy N.B.C.

so propose

$$\phi(x, t) = A_0 + \sum_{n=1}^{\infty} A_n e^{-n^2 \pi^2 t / Re} \cos(n \pi x)$$

basically

$$\phi = A_0 + A_1(\cosine 1) + A_2(\cosine 2) + \dots$$

what exp? $\Rightarrow \phi_n(x, t) = A_n e^{-\lambda t} \cos(n \pi x)$

we gotta satisfy $\Rightarrow \phi_t = \frac{1}{Re} \phi_{xx}$

$$\phi_t = -\lambda A_n e^{-\lambda t} \cos(n \pi x)$$

$$\phi_x = -n \pi A_n e^{-\lambda t} \sin(n \pi x)$$

$$\phi_{xx} = -n^2 \pi^2 A_n e^{-\lambda t} \cos(n \pi x)$$

put into heat eqn

$$-\lambda A_n e^{-\lambda t} \cos(n \pi x) = \frac{1}{Re} (-n^2 \pi^2 A_n e^{-\lambda t} \cos(n \pi x))$$

$$\therefore \boxed{\lambda = \frac{n^2 \pi^2}{Re}} \quad \text{so time plant} \Rightarrow \boxed{e^{-n^2 \pi^2 t / Re}}$$

heat eqn
forces the
time dependence

$\therefore$ cosine isn't enough to represent an arbitrary IC.

 $\rightarrow$ one cosine can't produce this

But a combi. $\Rightarrow A_0 + A_1 \cos(n \pi x) + \dots$

can approx. inc. compli. shapes

$A_0, A_1, A_2, \dots \rightarrow$ Fourier coeff $\rightarrow$ how much of each cosine mode do we need?

$$\phi(x, 0) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n \pi x)$$

if we know $\phi(x, 0) \rightarrow$ Fourier analysis tells us the A_n 's

Date: / /

⇒ This is when IC gets transferred

$$U(n,0) = U_0(n) \xrightarrow{\text{H.C.}} \phi(n,0) = n \exp\left[-\frac{Re}{2} \int_0^n U_0(\eta) d\eta\right]$$

Burg. IC $\rightarrow U(n,0) \xrightarrow{\text{H.C.}} \phi(n,0) \rightarrow$ Fourier decomp

$\rightarrow A_0, A_1, A_2, \dots$

Here $\Rightarrow t \uparrow$ for $n \uparrow \rightarrow n^2 \rightarrow$ high freq, much decay, much faster

(ex) $n=1 \rightarrow e^{-n^2 t/Re}$, $n=10 \rightarrow e^{-100n^2 t/Re}$

(ie) $n=10 \rightarrow$ disapp $\rightarrow$ much faster

physically $\rightarrow$ the heat eqn smooths the soln.

high freq / small scale vari. disp. first

↳ (ie) why diffn smooths things out.

$U_{nn} \rightarrow$ nonlinear transport $\rightarrow$ can create sharp structure

$v_{nn} \rightarrow$ diffn / smoothy $\rightarrow$ tries to smooth them

next step $\rightarrow$ recover "U"

Date: / /

$$\phi(n,t) \xrightarrow{\mu \text{ H.C.}} u = -\frac{2}{Re} \frac{\phi_n}{\phi}$$

(paper diff. Fourier series)

$$\rightarrow \phi u = - \sum_{n=1}^{\infty} A_n e^{-n^2 n^2 t/Re} n \sin(n\pi n)$$

then sub. into $\Rightarrow u = -\frac{2}{Re} \frac{\phi_n}{\phi}$

$$U(n,t) = \frac{2\pi}{Re} \sum_{n=1}^{\infty} A_n e^{-n^2 n^2 t/Re} n \sin(n\pi n)$$

$$A_0 + \sum_{n=1}^{\infty} A_n e^{-n^2 n^2 t/Re} \text{ (or } (n^2 n))$$

exact / ref. soln

(2) $U_0 \rightarrow \phi_0 \rightarrow A_n \rightarrow \phi(n,t) \rightarrow \phi_n \rightarrow U(n,t)$

(3) we need same problem def. as ref.

$$PDE + \text{domain} + IC + BC + v/Re$$

↳ if diff. comparing soln isn't meaningful

Date: / /

Madalabajo's
ex 2

$$\Rightarrow x \in [0, 1]$$

$$\& u(x, 0) = \sin(\pi x)$$

Raisi:

$$\Rightarrow x \in [-1, 1]$$

$$\& u(x=0) = -\sin(\pi x)$$

$$v = 0.01/\pi$$

why not we
this exact
method/
formula?

Can our PINN recover a known
soln

Can it solve problem where
an analytical soln isn't
available

while pt. of benchmark 1st PDE

H.C

$$u_t + u u_x = v u_{xx} \rightarrow u = -2v \frac{\phi_x}{\phi} \rightarrow \phi_t = v \phi_{xx}$$

F.S $\rightarrow \phi = A_0 + \sum A_n e^{-n^2 y/2 + t/n^2} \cos(n \pi x)$

H.C back trans $\rightarrow \phi \rightarrow u_{ref} \rightarrow u_{PINN}$ VS u_{ref} .